

Answers - Mendelian Randomization Exercise

1. Is there significant evidence of an association between LRRC37A2 and sensorineural hearing loss? [Yes](#)

2. Is there any evidence that some of the variants have horizontal pleiotropy? [Yes](#)

3. How many variants? [2](#)

4. What is the p-value, beta, and standard error before outlier removal? These are the results from the IVW test.

A. P-value: [6.03 x 10⁻⁴](#)

B. Beta: [9.04 x 10⁻³](#)

C. Standard Error: [2.56 x 10⁻³](#)

5. What is the p-value, beta, and standard error after outlier removal? Removal of the variants that display horizontal pleiotropy.

A. P-value: [5.37 x 10⁻⁵](#)

B. Beta: [1.01 x 10⁻²](#)

C. Standard Error: [2.41 x 10⁻³](#)

6. Why is the p-value smaller after outlier removal?

[The pleiotropic variants are outlier and increase the standard error, which in turn also makes the test statistic less significant. In this example, the beta is also larger after outlier removal, which also impacts the p-value, i.e., increases significance.](#)

7. Does an increase in LRRC37A2 protein expression levels increase or decrease the risk of developing sensorineural hearing loss? [Increases the risk of sensorineural hearing loss](#)

8. What is the potential underlying mechanism which is suggested by the results:

A) The risk of sensorineural hear loss is due to genetic variants, which in turn causes LRRC37A2 protein to be released in the blood, for example through hair cell death and/or damage? [No](#)

B) Genetic variants are modulating LRRC37A2 protein expression levels in the blood which modifies sensorineural hearing loss risk? [Yes](#)